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Direct Comparison of a Genetically Encoded Sensor and Small Molecule Indicator: Implications for Quantification of Cytosolic Zn^{2+}

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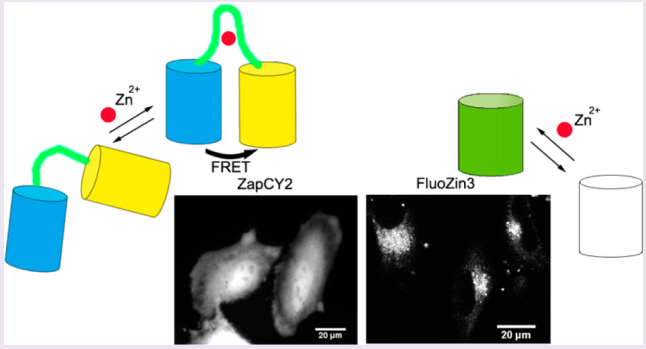
Supporting Information

ABSTRACT: Fluorescent sensors are powerful tools for visualizing and quantifying molecules and ions in living cells. A variety of small molecule and genetically encoded sensors have been developed for studying intracellular Zn^{2+} homeostasis and signaling, but no direct comparisons exist, making it challenging for researchers to identify the appropriate sensor for a given application. Here we directly compare the widely used small molecule probe FluoZin-3 and a genetically encoded sensor, ZapCY2. We demonstrate that, in contrast to FluoZin-3, ZapCY2 exhibits a well-defined cytosolic localization, provides estimates of Zn^{2+} concentration with little variability, does not perturb cytosolic Zn^{2+} levels, and exhibits rapid Zn^{2+} response dynamics. ZapCY2 was used to measure Zn^{2+} concentrations in 5 different cell types, revealing higher cytosolic Zn^{2+} levels in prostate cancer cells compared to normal prostate cells (although the total zinc is reduced in prostate cancer cells), suggesting distinct regulatory mechanisms.



Fluorescent sensors have become instrumental tools in cell biology, permitting researchers to visualize specific ions, molecules, and biochemical reactions in the complex environment of a cell. There are now over 120 genetically encoded fluorescent sensors for different targets[2] and hundreds of small molecule fluorescent indicators. Increasingly, such tools are used to quantify the amount of a particular entity, such as concentration of a metal ion or the kinetics of ion fluxes. A substantial challenge for users is how to rigorously compare sensors and evaluate quantitative estimates obtained using different probes, particularly given the vast array of probes with different molecular designs and biochemical and photophysical characteristics.[2−4] For example, there is considerable disagreement among the measurements of accessible Zn^{2+} in cells using different tools, with estimates varying by 3 orders of magnitude and ranging from 5 to 1000 pM.[5−14] One possible explanation for this variability could be the use of different cell types. Indeed the above estimates include red blood cells, leukemic cells, splenocytes, thymocytes, cardiomyocytes, rat PC-12 cells, HT-29 cells, HEK 293 cells, insulin secreting INS-1 (832/13) cells, hepatocytes, neurons, and HeLa epithelial cells. Because few studies compare a range of cell types, it is difficult to evaluate whether these differences reflect natural variation in cells with different Zn^{2+} regulatory mechanisms, or whether estimates vary due to the use of different probes and measurement conditions.

Zn^{2+} has been documented to influence many cellular processes, including proliferation, differentiation, and apoptosis,[15,16] as well as to inhibit enzymes such as caspases and phosphatases at concentrations ranging from picomolar to nanomolar.[1,17] Yet the precise concentration of accessible Zn^{2+}, its spatial distribution, and the conditions under which levels are dynamic are poorly understood. One of the major challenges in the field is that there have been no quantitative measurements of the distribution of free Zn^{2+} in a diversity of cell types and an assessment of how this picture changes with cellular state. Thus, the range of physiological concentrations remains ill-defined.

Here we establish a general framework for comparing and validating fluorescent probes for Zn^{2+}, enabling newly developed tools and probes from different classes (e.g., small molecule versus genetically encoded) to be benchmarked against existing tools. By comparing the widely used small molecule zinc sensor FluoZin-3 (K_d = 15 nM)[18] with a representative genetically encoded sensor ZapCY2 (K_d = 811 pM, n = 0.44),[12] we reveal that indeed different probes can yield substantially different results. Given the dizzying array of new probes and the increasing incidence with which they are

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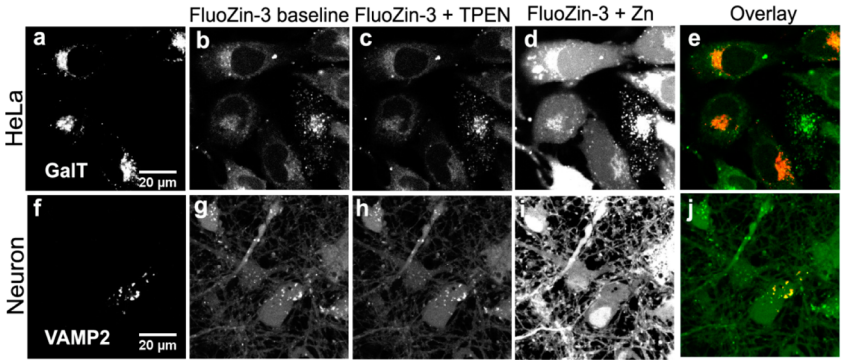


Figure 1. Localization of FluoZin-3 in HeLa cells and cultured cortical neurons. Fluorescence image of mCherry-GalT Golgi marker in HeLa (a) and mCherry-VAMP2 in neurons (f). FluoZin-3 staining at rest in HeLa (b) and neurons (g), after treatment with 100 μM TPEN (c, h), and after incubation with 100 μM $ZnCl_2$ and 5 μM pyithione (d, i). Overlay of Golgi (red) with FluoZin-3 (green) for HeLa (e) and VAMP2 (red) with FluoZin-3 (green) for neurons (j). Threshold of images at low zinc level (b, c and g, h) is adjusted lower than images for zinc saturation (d, i) to enable visualization with naked eyes. Images b and c are thresholded at the same level, and g and h are thresholded at the same level.

used to measure biological processes such a framework will be critically important for the scientific community, both probe developers and the end-users. Lastly, using the validated genetically encoded sensor ZapCY2 we compare the cytosolic Zn^{2+} levels in five cell types, revealing an increase in accessible cytosolic Zn^{2+} in a prostate cancer cell line compared to a normal prostate cell line.

RESULTS AND DISCUSSION

Establishing Subcellular Localization. There is growing evidence that the subcellular distribution of Zn^{2+} is inhomogeneous, with different concentrations in different organelles.[12,13,19,20] Therefore, when quantifying cellular Zn^{2+} levels it is essential to precisely define the identity of the compartment being measured. The location of genetically encoded sensors can be controlled by incorporation of a signal peptide into the sensor, and the sensor can be explicitly targeted to the cytosol by incorporation of a nuclear exclusion signal (Supplementary Figure S1). FluoZin-3 is a widely used small molecule Zn^{2+} indicator, employed in hundreds of publications. This membrane permeable dye is trapped inside cells upon cleavage of the acetoxymethyl (AM) ester by cellular esterases.[21] The localization of FluoZin-3 is not precisely defined, and the dye is at least partially sequestered into intracellular compartments. Moreover, the localization of the dye varies with cell type. Indeed, FluoZin-3 has been used as a cytosolic Zn^{2+} sensor[22−25] but has also been used as an indicator for vesicular Zn^{2+},[26,27] lysosomal Zn^{2+} in oxidative neurons,[28] and vesicles in HeLa cells when the expression of ZIP13 is reduced.[29]

In our hands in HeLa cells, FluoZin-3 partially co-localizes with a Golgi marker (Figure 1, Supplementary Figure S2a−c and Table S1, Pearson's coefficient = 0.60 ± 0.18) but is also present in the cytosol, as evidenced by the increase in cytosolic fluorescence upon treatment with Zn^{2+} (Figure 1 g). The high fluorescence signal in Golgi compared to other regions did not decrease upon zinc chelation with 100 μM TPEN (Figure 1e), suggesting that the high signal in Golgi was not due to high Zn^{2+} but rather to accumulation of the dye. In cultured cortical neurons and insulin-secreting MIN6 cells FluoZin-3 showed poor co-localization with a Golgi marker (Supplementary Figues S2 and Table S1, Pearson's coefficient = 0.41 ± 0.32 for neurons and 0.37 ± 0.40 for MIN6).

Instead, in neurons FluoZin-3 partially co-localized with the synaptic vesicle marker VAMP2 (Figure 1, Supplementary Figure S2g−i and Table S1, Pearson's coefficient = 0.66 ± 0.19). However, the dye was also present in the cytosol and nucleus as treatment with 5 μM pyrithione and 100 μM $ZnCl_2$ (Figure 1h) yielded bright fluorescent signals in both locations.

Quantifying Intracellular Zn^{2+}. Comparing cytosolic Zn^{2+} concentration ($[Zn^{2+}]$) in different cells or in cells of different diseased states is an important step in evaluating how resting Zn^{2+} levels and deviations from resting levels influence cellular function. The most precise estimates of $[Zn^{2+}]$ involve calibration of the sensor in the cellular milieu and determination of the percent saturation of the sensor in each individual cell, followed by conversion to $[Zn^{2+}]$. Figure 2 presents representative calibrations of ZapCY2 and FluoZin-3. Each trace represents an individual cell, and it is evident that even under resting conditions there is substantial variability in the FRET ratio (ZapCY2) or fluorescence intensity (FluoZin-3) from cell to cell. However, when the parameters of the *in situ* calibration were used to determine the percent saturation of the sensor, ZapCY2 exhibited a narrow distribution (26.7−28.3%, Figure 2a), whereas the variability was much greater for FluoZin-3 in HeLa cells (15.1−41.1%, Figure 2b).

Another important consideration for accurately comparing cellular $[Zn^{2+}]$ is the extent to which the sensors themselves perturb free ion levels. In particular, if the [sensor] inside cells is comparable to the buffered Zn^{2+} pool, the sensor could lead to depletion of free Zn^{2+}.[30] To test whether sensors perturb the free Zn^{2+} pool, the fractional saturation was measured as a function of [sensor]. For ZapCY2, the percent saturation of sensor was maintained at 23−32% in different cells and was not greatly affected by [sensor] (Figure 2c). Conversely for FluoZin-3, the percent saturation varied from 45% to 20% and decreased with increasing [FluoZin-3] (Figure 2d), indicating that estimates of free Zn^{2+} with FluoZin-3 depend on the amount of dye internalized.

Comparison of Intracellular Zn^{2+} in Different Cells. ZapCY2 was used to compare the free Zn^{2+} levels in 5 cell types under resting conditions. Full *in situ* calibrations were carried out for HeLa, RWPE1, LNCaP, MIN6, and primary cultured cortical neurons, and the resting FRET ratio was converted into a $[Zn^{2+}]$, yielding estimates of mean cytosolic Zn^{2+} of ∼60 pM for normal RWPE1 cells, ∼ 270 pM for cancerous LNCaP cells, ∼110 pM in MIN6, and 80 pM in neurons (Figure 2e). While

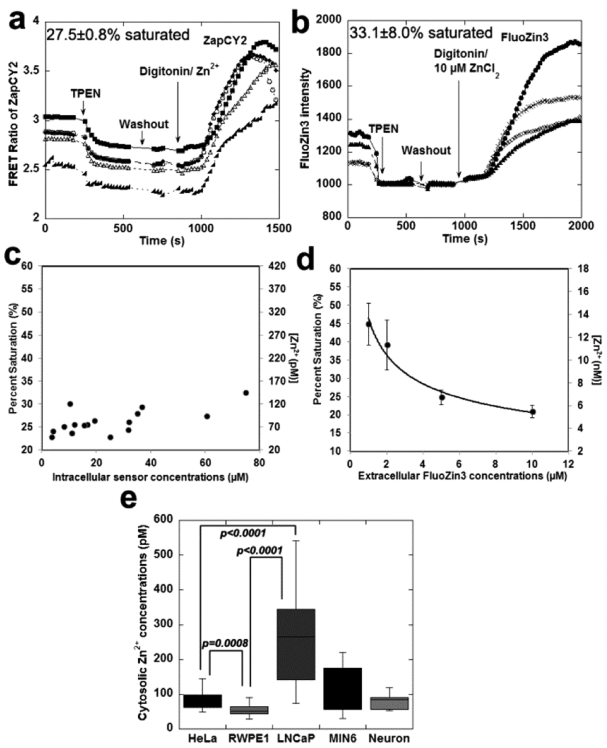


Figure 2. Reduced signal variability and zinc buffering of ZapCY2 compared to FluoZin-3. HeLa cells expressing ZapCY2 (a) or loaded with FluoZin-3 (b) and subjected to calibration. Percent saturation of ZapCY2 and estimated [Zn^{2+}] at rest as a function of intracellular [sensor] (c) and for FluoZin-3 as a function of the extracellular [FluoZin-3] used to treat cells (fit to an exponential decay) (d). Intracellular [FluoZin-3] is typically ∼100× higher than the extracellular concentration. Data are from 6 cells at each concentration. (e) Cytosolic [Zn^{2+}] in different cell types. Box represents middle quartiles, and whiskers represent upper and lower quartile (full statistics in Supplemenary Table S2).

there can be substantial variability in [Zn^{2+}] in some cell types (note LNCaP), on average, the cytosolic Zn^{2+} in cancerous LNCaP is significantly higher than normal RWPE1 ($P < 0.001$).

***In Vivo* Kinetic Response.** Lastly, we compared the temporal response of ZapCY2 and FluoZin-3 upon acute manipulation of intracellular Zn^{2+}. MIN6 cells were used to exploit their voltage-gated calcium channels (L-VGCC) that open upon depolarization and are permeable to Zn^{2+}.[31] When MIN6 cells were treated with Zn^{2+} under depolarization conditions, both sensors showed an increased signal and more rapid response compared to cells treated with $ZnCl_2$ alone (Figure 3a and b). In both conditions, ZapCY2 showed a quicker response compared to FluoZin-3. To rule out the possibility that this phenomenon was caused by the affinity difference (ZapCY2 is saturated at lower [Zn^{2+}] than FluoZin-3), MIN6 cells were treated with pyrithione and $ZnCl_2$ to ensure rapid influx of high levels of Zn^{2+}. Under these conditions, saturation of ZapCY2 occurred more rapidly than for FluoZin-3 (Figure 3c). One explanation for the kinetic difference could be that the [FluoZin-3] is much higher than [ZapCY2] in cells. To test this, we treated MIN6 cells with different [FluoZin-3] (0.5−10 μM) for 40 min and then perfused the cells with 5 μM pyrithione and 100 μM $ZnCl_2$. Cells with a lower amount of FluoZin-3 reached the maximal intensity faster than the cells with higher FluoZin-3 (Figure 3d). However the rate of entry was still slower than for ZapCY2. Computational simulations also provided supporting evidence that cells with higher [dye] display slower kinetics (Figure 3e).

In summary, ZapCY2 exhibited a number of distinct advantages when compared to FluoZin-3. First, the localization can be explicitly defined to be cytosolic, whereas FluoZin-3 reports on Zn^{2+} changes within the cytosol as well as membrane enclosed compartments, and that localization varies significantly with cell type (Figure 1 and Supplementary Figure S1). Hence interpretation of FluoZin-3 data is confounded by

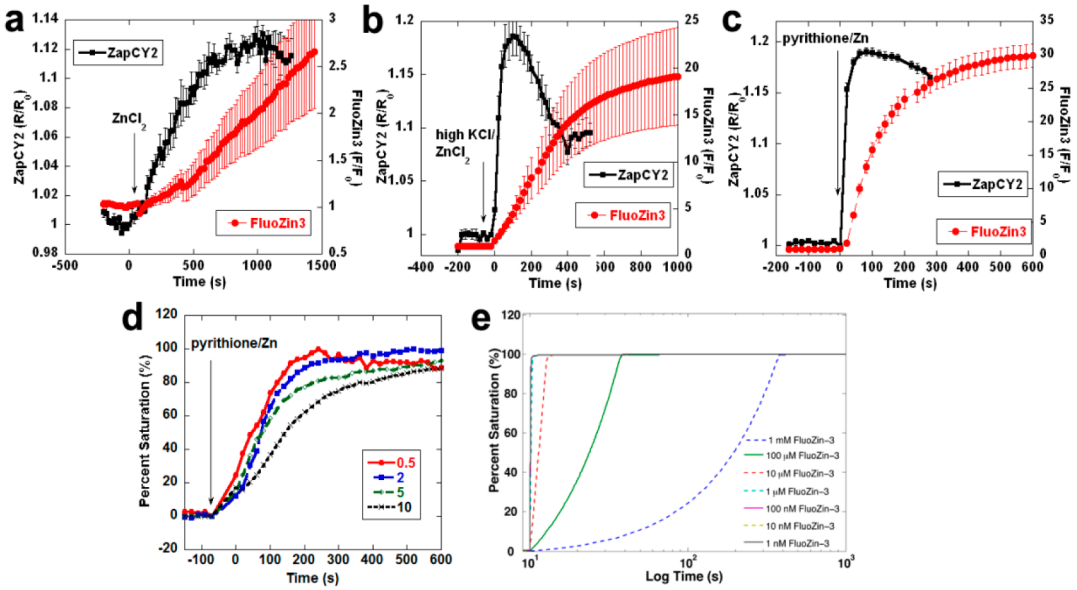
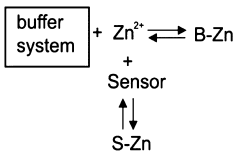


Figure 3. Comparison of the temporal response of FluoZin-3 and ZapCY2 in MIN6 cells. The increase in intracellular Zn^{2+} was induced by incubation with 100 μM $ZnCl_2$ (a), 50 mM KCl and 100 μM $ZnCl_2$ (b), or 5 μM pyrithione and 100 μM $ZnCl_2$ (c). MIN6 cells loaded with 0.5, 2, 5, or 10 μM FluoZin-3 were treated with pyrithione/100 μM $ZnCl_2$ (d). Data are reported as saturation percentage $[(F - F_0)/(F_{max} - F_0) \times 100]$. Each plot represents the average of at least five cells. (e) Simulation of Zn^{2+} binding kinetics of FluoZin-3. In this simulation model, higher concentrations of FluoZin-3 would take a longer time to reach the saturation. The error bars indicate SEM (standard error of the mean).

an inability to precisely define where the Zn^{2+} changes are occurring. This may explain why estimates of cytosolic Zn^{2+} using FluoZin-3[6,32] are typically higher than estimates using genetically encoded sensors,[12,13] particularly in cells containing vesicular Zn^{2+}. In addition, the accumulation of FluoZin-3 in subcellular compartments such as the Golgi leads to high fluorescence (Figure 1), which may be incorrectly interpreted as elevated Zn^{2+} unless an *in situ* calibration is performed.

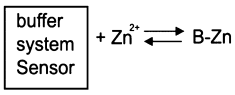
A second advantage is that in distinct contrast to FluoZin-3, ZapCY2 does not appear to perturb the resting free Zn^{2+} pool. Scheme 1 presents a simplified model of Zn^{2+} homeostasis and

Scheme 1



illustrates the potential of an exogenous sensor to perturb steady state Zn^{2+}. This scheme suggests that if the [sensor] approaches that of the buffer system, the sensor may deplete the cellular ion pool. In this work we measured the free Zn^{2+} as a function of [sensor] for ZapCY2 and FluoZin-3. We found that while free Zn^{2+} estimates based on ZapCY2 were independent of the intracellular [sensor] (Figure 2c), there was a strong dependence with increasing FluoZin-3, leading to a significant decrease in the measured fractional saturation (Figure 2d), suggesting depletion of the cellular ion pool. This is consistent with previous studies using FluoZin-3 in human colon cancer (HT-29) cells and neurons[6,30] and likely results from the fact that AM ester versions of Zn^{2+} indicators are typically concentrated at least 100-fold, yielding intracellular concentrations in the hundreds of micromolar.[6,30] The total buffered Zn^{2+} pool in mammalian cells has been estimated in the hundreds of micromolar,[6,33,34] so it is possible that the lower levels of ZapCY2 (1−10 μM) compared to the amount of internalized FluoZin-3 (hundreds of micromolar) are not sufficient to perturb the equilibrium and deplete the Zn^{2+} pool. However, another difference between the two probes is that measurements with genetically encoded sensors are performed days after introduction of the sensor into cells, whereas those with FluoZin-3 are performed within a few hours of loading the dye, due to extrusion of the dye over time. Perhaps long-term expression of ZapCY2 allows it to become part of the zinc buffer system, as represented in Scheme 2.

Scheme 2



Lastly, a final advantage of ZapCY2 is the rapid *in vivo* response compared to that of FluoZin-3 (Figure 3). We suspect that the relatively rapid response of ZapCY2 results from the fact that there is significantly less sensor within the cell, and hence saturation of the sensor occurs more quickly. This was supported by kinetic modeling showing that higher concentrations of sensor lead to slower kinetic responses. These data suggest that ZapCY2 might be a better tool for studying zinc uptake and efflux kinetics compared to FluoZin-3.

Given the wide range of estimates for resting free Zn^{2+} levels and the scarcity of studies that directly compare Zn^{2+} in a range of diverse cell types, we used ZapCY2 to compare relative Zn^{2+} levels in 3 types of epithelial cells: HeLa, normal prostate RWPE1, and cancerous prostate LNCaP, as well as neurons and a model pancreatic beta cell line (MIN6). We found no difference in cytosolic Zn^{2+} between HeLa, MIN6, and cortical neurons, but a significant decrease in RWPE1 and an increase in LNCaP. The higher level of free Zn^{2+} in the cytosol of LNCaP versus RWPE1 is particularly intriguing given that it is well established that *total* Zn^{2+} levels in prostate cancer cells are lower than normal cells and that Zn^{2+} content strongly correlates with metastasis.[35] Indeed, using inductively coupled mass spectrometry (ICP-MS) we verified that total cellular Zn^{2+} in LNCaP is 65% lower than in RWPE1 cells (Supplementary Figure S3). Thus, our study reveals the importance of defining Zn^{2+} levels in explicit subcellular locations and reveals for the first time that cytosolic Zn^{2+} is actually higher in the prostate cancer cell line, despite lower levels of total Zn^{2+}. It is interesting to speculate that high cytosolic Zn^{2+} in cancer cells may contribute to these cells being more refractory toward apoptosis, as Zn^{2+} can inhibit caspase 3 activity and repress apoptosis.[1] If this is the case, addition of a Zn^{2+} chelator might be used to induce apoptosis in cancer cells, an approach that has been used in the development of new agents for reducing tumor size in animal models.[36]

In conclusion, direct comparisons between ZapCY2 and FluoZin-3 showed that the genetically encoded sensor can overcome many limitations of this small molecule sensor when the experimental goal is to define Zn^{2+} in a precise location (such as the cytosol), minimize perturbation of the cellular ion pool, facilitate Zn^{2+} quantification, and report the kinetics of cellular Zn^{2+} dynamics.

■ METHODS

Details of the plasmids, culture, transfection conditions for all cells, method for quantifying sensor concentration in cells, and FluoZin3 staining co-localization analysis are provided in Supporting Information.

Cellular Zn^{2+} Measurements. Imaging experiments were performed on a Zeiss Axiovert 200 M inverted fluorescence microscope configured as previously described.[12] Images were collected with a 40X 1.3NA oil objective, using 5% light transmission and 300 ms exposure time. Images of ZapCY2 were background corrected by generating a region of interest (ROI) on a blank area of the coverslip and subtracting the fluorescence intensity of each channel, e.g., I_{FRET}(sample) − I_{FRET}(background), and I_{CFP}(sample) − I_{CFP}(background). The background corrected FRET and CFP images were used to calculate the FRET ratio (R, i.e., FRET/CFP). Images of FluoZin-3 were not background corrected because the resting signal was very low and close to the background.

Cells were imaged in phosphate-free HHBSS, pH = 7.4 to prevent precipitation of zinc. To carry out *in situ* calibrations, cells were treated with 150 μM cell-permeable zinc chelator *N,N,N′,N′*-tetrakis(1-pyridylmethyl) ethylenediamine (TPEN) to obtain the FRET ratio at minimum Zn^{2+} (R_{min}) and addition of 5 μM pyrithione and 10 or 100 μM $ZnCl_2$ to establish the FRET ratio at maximum Zn^{2+} (R_{max}).

Statistical Analysis. Statistical analysis was performed using the *t* test in the KaleidaGraph program.

■ ASSOCIATED CONTENT

Supporting Information

Additional methods and results. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

FRET, Förster resonance energy transfer; K_D', dissociation constant; [Zn^{2+}], free Zn^{2+} concentration; [sensor], sensor concentration

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